

PAPER CODE: BCE-C625
EVEN SEMESTER EXAMINATION (MAY-2026)
CLASS: B.TECH. SEMESTER: VI
PAPER TITLE: THEORY OF COMPUTATION:CLASSICAL
TO QUANTUM MODELS

Time: 3 hours

Max. Marks: 70

Note: Question Paper is divided into two sections: **A and B**. Attempt both the sections as per given instructions.

SECTION-A (SHORT ANSWER TYPE QUESTIONS)

Instructions: Answer any *five* questions in about 150 words each. Each question carries six marks.
 (5 x 6 = 30 Marks)

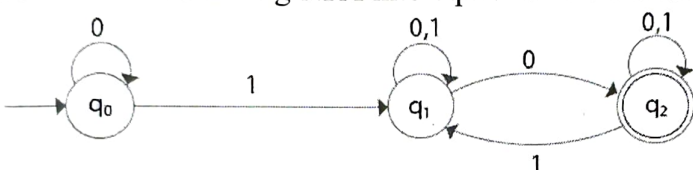
		CO	BL
Q-1	Construct a DFA over $\Sigma = \{0,1\}$ that accepts all strings ending with "01".	2	2
Q-2	Define Deterministic Finite Automaton (DFA). Write the formal 5-tuple representation of DFA.	3	3
Q-3	Differentiate between Moore Machine and Mealy Machine with suitable examples.	1	2
Q-4	Convert the following regular expression into NFA: $(a+b)^*ab$	3	3
Q-5	Write a context free grammar (CFG) for the language: $L = \{a^n b^n n \geq 1\}$	1	2
Q-6	What is ambiguity in context free grammar? Give one example of ambiguous grammar.	2	1
Q-7	Construct a leftmost derivation for the string $a^4 a b b b b^2$ using the grammar: $S \rightarrow aSb \mid ab$	5	4
Q-8	Define Pushdown Automata (PDA). What is the role of stack memory in PDA?	2	2
Q-9	Explain the basic structure of a Turing Machine with the help of a diagram.	3	3

Q-10	Differentiate between decidable and undecidable problems with examples.	1	2
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SECTION-B (LONG ANSWER TYPE QUESTIONS)

Instructions: Answer any *four* questions in detail. Each question carries 10 marks.

(4 x 10 = 40 Marks)

		CO	BL
Q-1	Construct a DFA that accepts all binary strings containing an even number of 0's and an odd number of 1's. Also draw the transition table and explain the working of the automaton.	1	3
Q-2	Convert the following NFA into equivalent DFA. Show all steps. 	3	4
Q-3	Find the regular expression corresponding to the following language: "All strings over {a,b} that start with 'a' and end with 'bb'." Also design the equivalent finite automaton.	2	3
Q-4	Eliminate left recursion from the following grammar and construct the FIRST and FOLLOW sets: $A \rightarrow Aa A\beta \gamma$	3	4
Q-5	Convert the following CFG into Chomsky Normal Form (CNF): $S \rightarrow aAB b$ $A \rightarrow aA a$ $B \rightarrow bB b$	2	3
Q-6	Design a Pushdown Automaton for the language: $L = \{wcw^R \mid w \in \{a,b\}^*\}$ Explain each transition with stack operation.	3	4
Q-7	Explain Chomsky hierarchy of languages in detail.	1	6
Q-8	What is the Halting Problem? Prove that the Halting Problem is undecidable using diagonalization or contradiction method.	2	6